GBBO Analysis

v2

Konnor Herbst

Table of contents

Introduction

We may be interested in questions like: *What actually matters in the tent? Does it differ for overall winner, Star Baker, and elimination? Who were the strongest contenders?*

Naive descriptive statistics are useful, but they only get us so far here. The structure of *The Great British Bake Off* is longitudinal: each baker contributes repeated performances until they are eliminated, and each episode defines a new risk set of bakers still eligible for Star Baker or elimination. A useful analysis should respect that structure, rather than relying only on marginal descriptive statistics that may ignore these subtleties and lead to misleading conclusions.

We build four main models:

1. **Elimination model** The outcome is whether a baker is eliminated in a given episode.
2. **Star Baker model** The outcome is whether a baker wins Star Baker in a given episode.
3. **Baker strength / ability model** The outcome is a baker's underlying performance level, interpreted through baker-specific random effects. This gives a model-based estimate of how far each baker is from the average baker, after accounting for repeated performances and episode-level structure.
4. **Running win probability model** The outcome is each baker's updated probability of winning the overall competition, recalculated sequentially after each episode using the information available up to that point.

These models have different targets. The elimination and Star Baker models are *judging behavior models*: they ask how episode-level performances are associated with being eliminated or winning Star Baker among the bakers at risk in that episode. The baker strength model is a *latent ability model*: it estimates persistent baker-level performance after accounting for repeated observations and episode context. The running win probability model is a *sequential prediction model*: it tracks how each baker's probability of winning changes as the season unfolds.

The main predictors available across these models are episode-level signature performance, showstopper performance, and technical rank. Signature and showstopper performance are scored using taste, appearance, and bake components, each coded on a $-1, 0, 1$ scale. Technical performance is represented by within-episode technical rank.

Because judging criteria and stakes may change over the course of a season, these effects can be allowed to vary by round through interactions with round.

Descriptive/Inferential Models vs Predictive

We will use the judging behavior models for both inferential and predictive purposes - we will fit them first using all episodes of interest for inference then we will fit them using LOOCV and sequential CV (trained on all preceding episodes) for prediction to evaluate performance

Judging Models

We have 9 available seasons/series,

n_rows	n_series	n_episodes	n_bakers
678	9	90	108

For the Star Baker and elimination outcomes, the relevant comparison is within episode. In each episode, the judges choose one Star Baker and, in most non-final episodes, one eliminated baker from the bakers currently competing. This motivates a conditional logit model with episode-specific strata. The episode stratum restricts comparisons to the appropriate risk set, rather than comparing bakers across different series or episodes where they were never actually competing against each other.

We exclude the final round, Round 10, from both the Star Baker and elimination models. In the final, "Star Baker" is effectively the overall winner, and there is no ordinary weekly elimination decision. For the elimination model, we also exclude episodes with zero or two eliminations, so that each retained risk set has exactly one eliminated baker. The same restriction is used during cross-validation.

A baseline frequentist conditional logit model assumes a shared judging rule across episodes and seasons. That is, the effects of signature, showstopper, and technical performance are assumed to be the same at every point in the competition. For baker i in episode j, write the underlying model as

$$\operatorname{logit} \Pr(Y_{ij} = 1) = \alpha_j + X_{ij}^\top \beta,$$

where Y_{ij} indicates whether baker i is selected in episode j, X_{ij} contains the signature, showstopper, and technical performance variables, and α_j is an episode-specific intercept.

The episode intercept α_j represents the baseline selection level for episode j. It is shared by all bakers in the same episode and captures episode-level differences in baseline selection probability, such as the mechanical difference between choosing one baker out of twelve versus one baker out of three.

In the conditional logit model, we compare bakers only within the same episode. Equivalently, we condition on the observed number of selected bakers in each episode. For a one-event episode, the conditional probability that baker i is selected is

$$\Pr\left(Y_{ij} = 1 \mid \sum_{\ell \in \mathcal{R}_j} Y_{\ell j} = 1\right) = \frac{\exp(X_{ij}^\top \beta)}{\sum_{\ell \in \mathcal{R}_j} \exp(X_{\ell j}^\top \beta)},$$

where $\mathcal{R}_j$ is the episode-specific risk set.

The episode intercept α_j cancels from this conditional probability:

$$\frac{\exp(\alpha_j + X_{ij}^\top \beta)}{\sum_{\ell \in \mathcal{R}_j} \exp(\alpha_j + X_{\ell j}^\top \beta)} = \frac{\exp(X_{ij}^\top \beta)}{\sum_{\ell \in \mathcal{R}_j} \exp(X_{\ell j}^\top \beta)}.$$

Thus, α_j is allowed to be arbitrary but is not directly estimated. The conditional likelihood across episodes is

$$L_c(\beta) = \prod_j \prod_{i \in \mathcal{R}_j} \left[\frac{\exp(X_{ij}^\top \beta)}{\sum_{\ell \in \mathcal{R}_j} \exp(X_{\ell j}^\top \beta)} \right]^{Y_{ij}}$$

Since each retained episode has exactly one selected baker, this can also be written as

$$L_c(\beta) = \prod_j \frac{\exp(X_{i_j j}^\top \beta)}{\sum_{\ell \in \mathcal{R}_j} \exp(X_{\ell j}^\top \beta)},$$

where i_j is the selected baker in episode j.

Because we condition on the episode-specific event totals, this is a likelihood for β alone, not the full unconditional likelihood for (α, β). Algebraically, it is the same likelihood as a stratified Cox proportional hazards partial likelihood with constant artificial event time and one event per stratum. In this application, the Cox model is only a computational device; the substantive interpretation is as a within-episode conditional choice model.

In this setup, β estimates how within-episode differences in signature, showstopper, and technical performance are associated with selection as Star Baker or elimination.

Covariates / Predictors

Because the Star Baker and elimination models are fit within episode, the predictors are interpreted as within-episode differences among bakers competing in the same risk set.

The main predictors are:

1. **Signature sum**
   The baker's total signature score in that episode. This is the sum of the signature bake, flavor, and appearance components. Each component is scored on a $-1, 0, 1$ scale, so the total signature score ranges from -3 to 3.

2. **Showstopper sum**
   The baker's total showstopper score in that episode. This is the sum of the showstopper bake, flavor, and appearance components. Each component is scored on a $-1, 0, 1$ scale, so the total showstopper score ranges from -3 to 3.
3. **Technical rank**
   The baker's rank in the technical challenge for that episode. Lower values indicate better performance: rank 1 is best, rank 2 is second best, and so on.

Signature sum and showstopper sum are on the same scale, so their effects are directly comparable. Technical rank is different: it is a within-episode ordinal placement, not a component score. A one-unit increase in technical rank means moving one place worse in the technical challenge.

Main Effects Only Models for Star baker and Elimination

We can use the clogit function in the survival package of R to fit both the elimination and star baker models. Starting with the elimination model

```
Call:
coxph(formula = Surv(rep(1, 579L), eliminated) ~ sig_sum + showstopper_sum +
    tech_rank + strata(episode), data = judging_elim, method = "exact")

  n= 579, number of events= 73

                   coef exp(coef) se(coef)     z Pr(>|z|)
sig_sum         -0.7185    0.4875   0.1263 -5.69  1.3e-08 ***
showstopper_sum -0.7812    0.4578   0.1194 -6.54  6.0e-11 ***
tech_rank        0.4402    1.5529   0.0859  5.13  2.9e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                exp(coef) exp(-coef) lower .95 upper .95
sig_sum             0.487      2.051     0.381     0.624
showstopper_sum     0.458      2.184     0.362     0.579
tech_rank           1.553      0.644     1.312     1.838

Concordance= 0.929  (se = 0.018 )
Likelihood ratio test= 151  on 3 df,   p=<2e-16
Wald test            = 58.3  on 3 df,   p=1e-12
Score (logrank) test = 126  on 3 df,   p=<2e-16
```

For elimination, better signature, showstopper, and technical performances are all associated with lower odds of being eliminated. A one-unit increase in signature score is associated with an odds ratio of 0.487, corresponding to a 51.3% reduction in the odds of elimination, holding the other predictors fixed within episode. A one-unit increase in showstopper score is associated with an odds ratio of 0.458, corresponding to a 54.2% reduction in the odds of elimination.

Signature and showstopper are on the same scale: both are summed component scores, with bake, flavor, and appearance contributing to the total. Therefore, their odds ratios are directly comparable as one-unit increases in judged performance totals.

The technical variable is different. Here, `tech_rank` is still a rank, not a score. Lower technical rank means better technical performance: rank 1 is best, rank 2 is next best, and so on. The model reports an odds ratio of 1.553 for a one-rank increase, meaning that moving one place worse in the technical challenge is associated with 55.3% higher odds of elimination. Equivalently, moving one place better in technical rank is associated with an odds ratio of

$$\frac{1}{1.553} \approx 0.644,$$

or about 35.6% lower odds of elimination, holding signature and showstopper scores fixed within the same episode.

Because `tech_rank` is measured on a different scale from `sig_sum` and `showstopper_sum`, its odds ratio should not be compared directly to theirs as if all three represented the same kind of one-unit performance improvement. A one-unit improvement in signature or showstopper score means one additional judged component point. A one-unit improvement in technical rank means moving one placement higher in the within-episode technical ranking.

A more meaningful comparison is to look at multi-unit improvements. For example, a two-point increase in signature score is associated with an elimination odds ratio of approximately

$$0.487^2 \approx 0.237,$$

or about 76.3% lower odds of elimination. A two-point increase in showstopper score is associated with an odds ratio of approximately

$$0.458^2 \approx 0.210,$$

or about 79.0% lower odds of elimination. By contrast, improving by two technical ranks is associated with an odds ratio of approximately

$$0.644^2 \approx 0.415,$$

or about 58.5% lower odds of elimination.

Thus, better technical placement clearly reduces elimination risk, but a few points of improvement in signature or showstopper scores correspond to larger changes in elimination odds than moving a few places up in the technical ranking.

Performance improvement	Odds ratio for elimination	Approximate reduction in odds
+1 signature point	0.487	51.3% lower
+2 signature points	$0.487^2 \approx 0.237$	76.3% lower
+1 showstopper point	0.458	54.2% lower
+2 showstopper points	$0.458^2 \approx 0.210$	79.0% lower
1 technical rank better	$1/1.553 \approx 0.644$	35.6% lower
2 technical ranks better	$0.644^2 \approx 0.415$	58.5% lower

overall implying that signature and shopwstopper are similarly important and technical rank increase is less so. However it is important to note that we did not allow effect modification via itneration by round that may allow importance to shift across rounds.

now for the star baker model

```
Call:
coxph(formula = Surv(rep(1, 678L), star_baker) ~ sig_sum + showstopper_sum +
    tech_rank + strata(episode), data = judging_sb, method = "exact")

  n= 678, number of events= 90

                   coef exp(coef) se(coef)     z Pr(>|z|)
sig_sum          0.7160    2.0462   0.1416  5.06  4.3e-07 ***
showstopper_sum  1.7225    5.5986   0.2726  6.32  2.6e-10 ***
tech_rank       -0.3958    0.6732   0.0792 -4.99  5.9e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                exp(coef) exp(-coef) lower .95 upper .95
sig_sum             2.046      0.489     1.550     2.701
showstopper_sum     5.599      0.179     3.282     9.552
tech_rank           0.673      1.486     0.576     0.786

Concordance= 0.929  (se = 0.016 )
Likelihood ratio test= 205  on 3 df,   p=<2e-16
Wald test            = 58.7  on 3 df,   p=1e-12
Score (logrank) test = 134  on 3 df,   p=<2e-16
```

For Star Baker, better signature, showstopper, and technical performances are all associated with higher odds of winning Star Baker. A one-unit increase in signature score is associated with an odds ratio of 2.046, meaning the odds of winning Star Baker are about 104.6 higher, holding the other predictors fixed within episode. A one-unit increase in showstopper score is associated with an odds ratio of 5.599, meaning the odds are about 459.9 higher.

Signature and showstopper are on the same scale: both are summed component scores, with bake, flavor, and appearance contributing to the total. Therefore, their odds ratios are directly comparable as one-unit increases in judged performance totals. On this scale, showstopper performance is much more strongly associated with Star Baker selection than signature performance. A one-point showstopper improvement is associated with more than five times the odds of winning Star Baker, while a one-point signature improvement is associated with about twice the odds.

The technical variable is different. Here, `tech_rank` is still a rank, not a score. Lower technical rank means better technical performance: rank 1 is best, rank 2 is next best, and so on. The model reports an odds ratio of 0.673 for moving one rank worse. Equivalently, moving one technical rank better is associated with an odds ratio of

$$\frac{1}{0.673} \approx 1.486,$$

or about 48.6 higher odds of winning Star Baker, holding signature and showstopper scores fixed within the same episode.

Moving two technical ranks better corresponds to an odds ratio of approximately

$$1.486^2 \approx 2.208,$$

or about 120.8 higher odds of winning Star Baker.

A two-point increase in signature score corresponds to an odds ratio of approximately

$$2.046^2 \approx 4.186,$$

or about 318.6 higher odds of winning Star Baker. A two-point increase in showstopper score corresponds to an odds ratio of approximately

$$5.599^2 \approx 31.349,$$

or about 3034.9 higher odds.

Again, technical rank should not be compared directly to signature and showstopper as if all three predictors were on the same scale. Technical rank is an ordinal placement, while signature and showstopper are summed judged component scores. Still, directionally aligned comparisons are useful: improving technical placement helps, but showstopper performance dominates Star Baker selection in this baseline model. A one-point showstopper improvement is associated with a much larger increase in Star Baker odds than moving one or even two places up in the technical ranking.

These comparisons summarize the baseline Star Baker model:

Performance improvement	Odds ratio for Star Baker	Approximate increase in odds
+1 signature point	2.046	104.6% higher
+2 signature points	$2.046^2 \approx 4.186$	318.6% higher
+1 showstopper point	5.599	459.9% higher
+2 showstopper points	$5.599^2 \approx 31.349$	3034.9% higher
1 technical rank better	$1/0.673 \approx 1.486$	48.6% higher
2 technical ranks better	$1.486^2 \approx 2.208$	120.8% higher

Again an important limitation of this baseline model is that these effects are assumed to be constant across rounds. The model treats a one-point improvement in signature, a one-point improvement in showstopper, or a one-rank improvement in technical placement as having the same association with Star Baker selection in Round 1 as in Round 9. That is likely too rigid, so we also fit round-interaction models to allow the importance of each performance component to change over the course of the season.

For this reason we include an interaction by round for both models (elim and star baker) below

Interacted (main effects + round interactions) Models for Star Baker and Elimination

Importantly below I center (zero) at round 1 so that main level coefficients have the interpretation as the effects at round 1

We can also allow the criteria to interact with round as we move along, usiing a round interaction by component, here the round main effect is not estimable because it is constant within episode because nesting - removing it from the model is not a violation of the hierarchical variables principle because the strata by episode is giving each episode its own baseline risk and the main round effect is absorbed into this, (alos at least one of the two in two way itneractions do have fit main effects)

```
Call:
coxph(formula = Surv(rep(1, 579L), eliminated) ~ sig_sum + showstopper_sum +
    tech_rank + sig_sum:round_c + showstopper_sum:round_c + tech_rank:round_c +
```

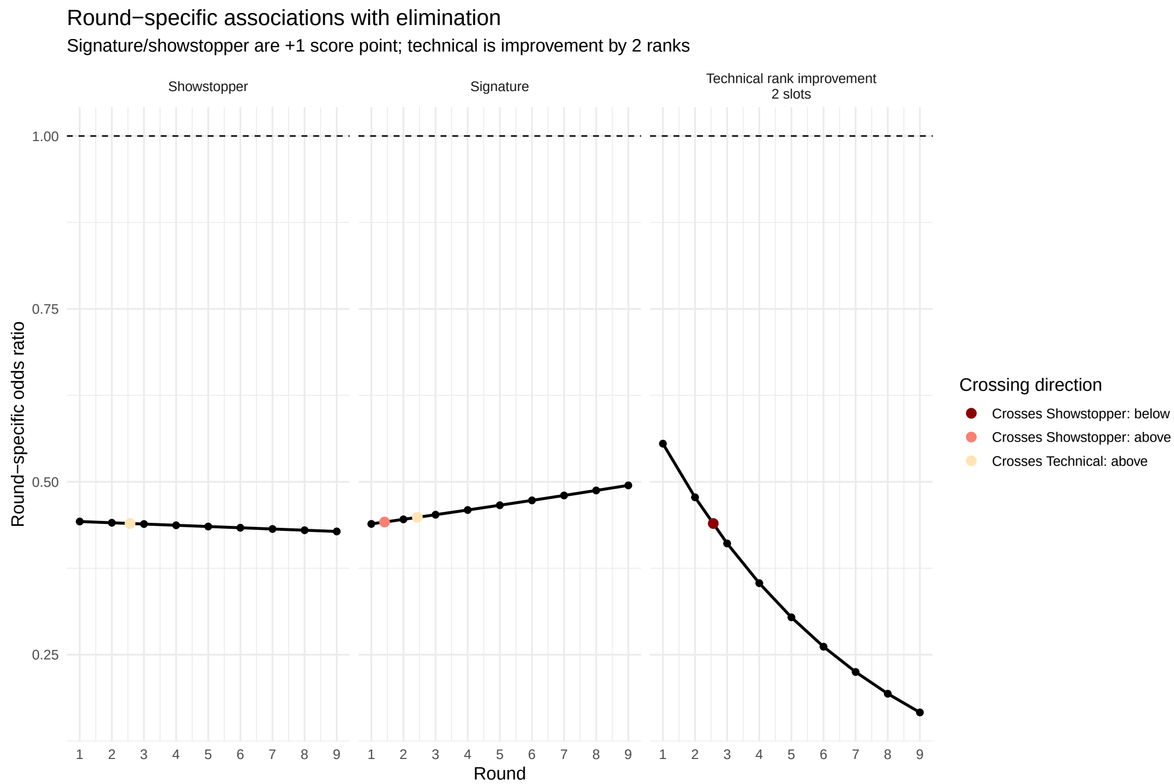
```
    strata(episode), data = judging_elim, method = "exact")

  n= 579, number of events= 73

                             coef exp(coef) se(coef)     z Pr(>|z|)    
sig_sum                  -0.82304   0.43909  0.23924 -3.44  0.00058 ***
showstopper_sum          -0.81516   0.44257  0.21449 -3.80  0.00014 ***
tech_rank                 0.29427   1.34214  0.10844  2.71  0.00665 ** 
sig_sum:round_c           0.01493   1.01505  0.05338  0.28  0.77966    
showstopper_sum:round_c -0.00412   0.99589  0.04734 -0.09  0.93065    
tech_rank:round_c         0.07526   1.07816  0.03908  1.93  0.05417 .  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                        exp(coef) exp(-coef) lower .95 upper .95
sig_sum                     0.439      2.277     0.275     0.702
showstopper_sum             0.443      2.260     0.291     0.674
tech_rank                   1.342      0.745     1.085     1.660
sig_sum:round_c             1.015      0.985     0.914     1.127
showstopper_sum:round_c     0.996      1.004     0.908     1.093
tech_rank:round_c           1.078      0.928     0.999     1.164

Concordance= 0.915  (se = 0.02 )
Likelihood ratio test= 158  on 6 df,   p=<2e-16
Wald test            = 55  on 6 df,   p=5e-10
Score (logrank) test = 128  on 6 df,   p=<2e-16
```



here it looks like sometime shortly around round 3 increasing in two slots of technical rank becomes more important than increasing a single component of flavor/looks/bake in showstopper or signature

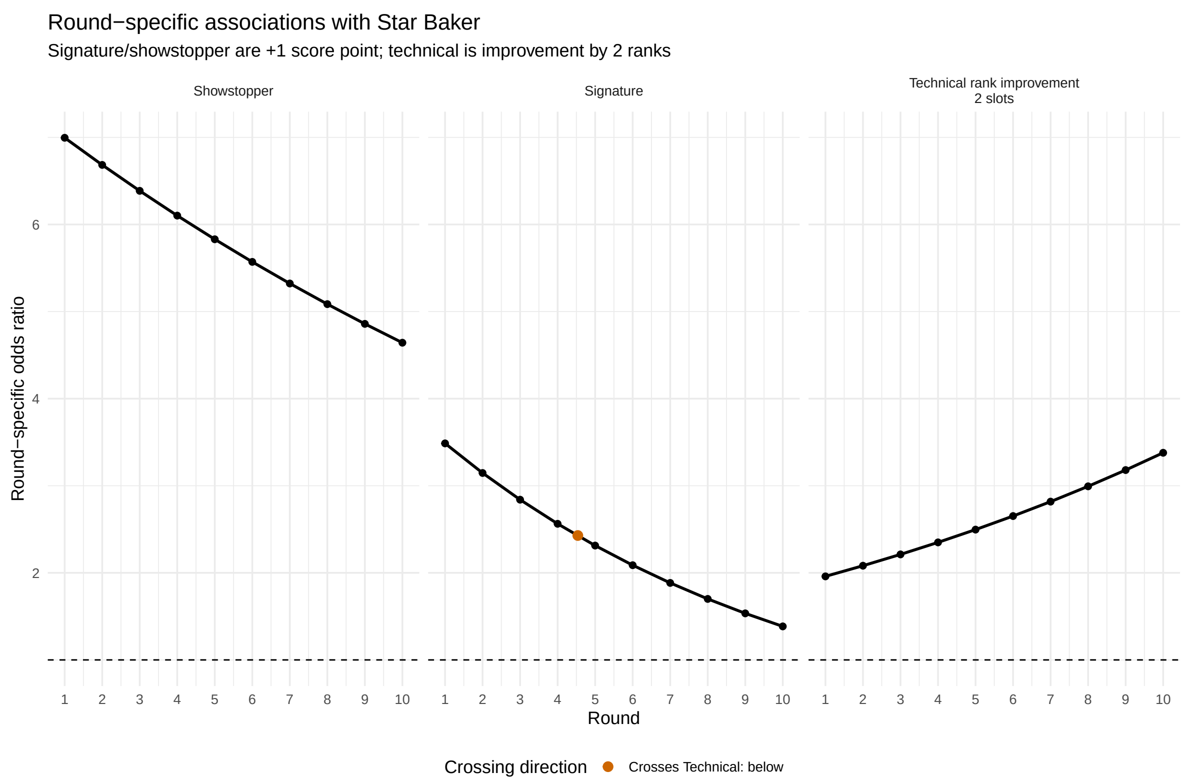
```
Call:
coxph(formula = Surv(rep(1, 678L), star_baker) ~ sig_sum + showstopper_sum +
    tech_rank + sig_sum:round_c + showstopper_sum:round_c + tech_rank:round_c +
    strata(episode), data = judging_sb, method = "exact")

  n= 678, number of events= 90

                           coef exp(coef) se(coef)     z Pr(>|z|)
sig_sum                  1.2489    3.4864   0.3585  3.48  0.00049 ***
showstopper_sum          1.9453    6.9955   0.5594  3.48  0.00051 ***
tech_rank               -0.3363    0.7144   0.1210 -2.78  0.00544 **
sig_sum:round_c         -0.1026    0.9025   0.0620 -1.66  0.09780 .
showstopper_sum:round_c -0.0456    0.9555   0.0999 -0.46  0.64836
tech_rank:round_c       -0.0303    0.9702   0.0329 -0.92  0.35786
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
                         exp(coef) exp(-coef) lower .95 upper .95
sig_sum                      3.486      0.287     1.727     7.039
showstopper_sum              6.995      0.143     2.337    20.940
tech_rank                    0.714      1.400     0.564     0.906
sig_sum:round_c              0.903      1.108     0.799     1.019
showstopper_sum:round_c      0.955      1.047     0.786     1.162
tech_rank:round_c            0.970      1.031     0.910     1.035

Concordance= 0.935  (se = 0.015 )
Likelihood ratio test= 210  on 6 df,   p=<2e-16
Wald test            = 56.8  on 6 df,   p=2e-10
Score (logrank) test = 135  on 6 df,   p=<2e-16
```



Here we have the effect of moving up two slots in the technical rank slowly increassign over time and trading importance with increasing a componbent of flavor/looks/bake in the signature around round 4-5 but the increasing showstopper components is overwhelmingly the most important here it seems.

Star-baker selection appears strongly associated with high performance in the showstopper while over time better performance in the technical is more protective against elimnation. This is consistent with star baker being a high-end performance award, where standing out matters, while elimination is a lower-tail selection problem, where the key issue is avoiding one of the weakest overall performances in the episode - most clearly found in low technical rank.

Just out of curiosity we check what proportion of those that scored highest in the showstopper sum win star baker, we filter by each round by the best showstopper scores required and see how many of those groupings include star baker,

n_episodes	n_contained_star_baker	prop_contained_star_baker
90	80	0.889

that's pretty high just from looking at who scored the best in each episodes showstopper!!!! That is just looking at their showstopper we can almost predict the group the star baker is in almost 90% of the time!

All this is evidence that

- Star Baker is a high-end award: it rewards standing out.
- For elimination, Technical performance appears most consequential in later rounds. Because elimination is a lower-tail selection problem, the model is asking which baker falls into the weakest part of the episode-specific risk set. As the risk set shrinks, poor Technical placement becomes harder to hide and can be especially damaging when Signature and Showstopper scores do not clearly separate another baker as weaker.

We can get a sense to see if the signature and showstopper perfomrance is more tight in latter rounds by looking at SD and IQR,

round	n_episodes	mean_n_bakers	mean_sig_sd	mean_sig_range	mean_show_sd	mean_show_range
1	9	12.00	1.79	5.11	1.76	4.78
2	9	11.11	1.79	4.89	1.54	4.11
3	9	9.89	1.58	4.67	1.86	5.22
4	9	9.00	1.64	4.33	1.87	5.33
5	9	8.33	1.70	4.22	1.62	4.56
6	9	7.00	1.96	5.00	1.90	4.78
7	9	6.00	1.60	4.00	1.66	4.11
8	9	5.00	1.74	4.11	1.66	4.00
9	9	4.00	1.27	2.78	1.44	3.11

looks like it a little bit, so the field narrows in terms of sig and showstopper (what they can porep for) and the elimination decision is based on the technical.

If I was going to track what mattered most over a season to coming out as the final winner this all seems to indicate that it would be signature performance until about round 3, then technical until round 9

(Interacted) Model Prediction Evaluation

I also evaluate how well the conditional logit models for star baker and elimination predict the selected baker within each episode (when trained on all other episodes). We see then that the models are not only descriptive; they have some ability to identify likely Star Baker and elimination candidates from the episode-specific risk set.

The prediction task is difficult because each episode has a changing set of bakers. Early episodes include many possible candidates, while later episodes include only a few. For each episode, the model assigns each baker a predicted probability, ranks the bakers from most to least likely to be selected, and then checks where the actual Star Baker or eliminated baker falls in that ordering.

Prediction is evaluated in two related ways. First, I use episode-level ranking metrics:

- top-1 accuracy: whether the actual selected baker has the highest predicted probability;
- top-2 and top-3 accuracy: whether the actual selected baker is among the model's top two or top three candidates;
- mean log loss: whether the model assigns high probability to the actual selected baker;
- median actual rank: where the true Star Baker or eliminated baker falls in the model's predicted ordering;
- mean actual probability: the model-assigned probability for the baker who was actually selected.

Second, I also summarize the same predictions as a binary classification problem over baker-episode rows. In this version, the top-ranked baker in each episode is treated as the model's positive prediction, while all other bakers are treated as negative predictions. I report the resulting confusion matrices, sensitivity, specificity, positive predictive value, AUROC, and AUPRC.

These binary metrics should be interpreted carefully. This is not an ordinary independent binary classification task because each episode has exactly one selected baker and the model is forced to rank bakers within that episode. Under the top-1 classification rule, sensitivity and positive predictive value are closely tied to top-1 accuracy: the model gets one positive prediction per episode, and there is one true positive per episode. Specificity is usually high because most baker-episode rows are non-selected bakers.

AUROC and AUPRC provide threshold-free summaries of whether selected bakers tend to receive higher predicted probabilities than non-selected bakers. However, the ranking metrics remain the most directly relevant evaluation criteria because the actual prediction problem is episode-specific: among the bakers still competing in a given week, who is most likely to be selected?

LOOCV

I use leave-one-episode-out cross-validation as the main prediction check. For each episode, the model is refit on all other episodes and then used to predict the held-out episode. Since the conditional logit model conditions on the episode-specific risk set, the held-out episode does not need an estimated stratum intercept. I use the fitted component coefficients to compute a linear predictor for each baker in the held-out episode, then renormalize those values into within-episode predicted probabilities.

The leave-one-episode-out results should still be interpreted as within-data-set validation. The held-out episodes come from the same overall collection of series, judges, scoring conventions, and data-processing choices. A stricter prediction task would be leave-one-series-out cross-validation, which would ask whether the estimated judging patterns generalize to an entirely unseen season.

n_episodes	top1_acc	top2_acc	top3_acc	sensitivity	specificity	ppv	auroc	auprc
90	0.689	0.889	0.989	0.689	0.952	0.689	0.945	0.701
73	0.616	0.822	0.945	0.616	0.945	0.616	0.923	0.599

- AUROC has a direct ranking interpretation: it is the probability that the model assigns a higher predicted probability to a randomly selected true event case than to a randomly selected non-event case.
  - For Star Baker, AUROC is the probability that an actual Star Baker is ranked above a non-Star Baker by predicted Star Baker probability.
  - For elimination, AUROC is the probability that an eliminated baker is ranked above a non-eliminated baker by predicted elimination probability.
- AUPRC summarizes the precision-recall tradeoff.
  - Precision asks: among the bakers the model assigns high predicted probability, how many were actually selected?
  - Recall asks: among the bakers who were actually selected, how many did the model successfully flag?
  - This is especially relevant here because each episode has only one Star Baker and usually one eliminated baker, so the event class is rare within each weekly risk set.

We also particularly look at how the model performs in the final stage - given those last three contestants

n_finals	n_correct	percent_right
9	6	66.7

outcome	Prediction	Truth	n
Star Baker	yes	yes	62
Star Baker	no	yes	28
Star Baker	yes	no	28
Star Baker	no	no	560
Elimination	yes	yes	45
Elimination	no	yes	28
Elimination	yes	no	28
Elimination	no	no	478

Table 9: Leave-one-episode-out confusion matrices

outcome	Prediction	Truth	n
Star Baker	yes	yes	62
Star Baker	no	yes	28
Star Baker	yes	no	28
Star Baker	no	no	560
Elimination	yes	yes	45
Elimination	no	yes	28
Elimination	yes	no	28
Elimination	no	no	478

here leave one out cross validation actualyl performs comparably well to naive!

Sequential CV

Sequential CV doesn't make the model suffer too much it seems!!!

Table 10: Sequential episode cross-validation metrics

outcome	top1_acc	top2_acc	top3_acc	sens	spec	ppv	auroc	auprc
Star Baker	0.675	0.912	0.988	0.675	0.950	0.675	0.950	0.739
Elimination	0.571	0.841	0.921	0.571	0.938	0.571	0.897	0.538

Table 11: Sequential episode cross-validation confusion matrices

outcome	Prediction	Truth	n
Star Baker	yes	yes	54
Star Baker	no	yes	26
Star Baker	yes	no	26
Star Baker	no	no	497
Elimination	yes	yes	36
Elimination	no	yes	27
Elimination	yes	no	27
Elimination	no	no	405

Table 12: LOOCV versus sequential episode cross-validation

cv_scheme	outcome	top1_acc	top2_acc	top3_acc	auroc	auprc
LOOCV	Star Baker	0.689	0.889	0.989	0.945	0.701
LOOCV	Elimination	0.616	0.822	0.945	0.923	0.599
Sequential CV	Star Baker	0.675	0.912	0.988	0.950	0.739
Sequential CV	Elimination	0.571	0.841	0.921	0.897	0.538

Model for baker's latent ability

We track each baker's scores across judged categories and stack them into a single long-format score outcome. Rather than fitting separate models for each score type, we model these scores jointly using a linear mixed-effects model. The model includes fixed effects for series and component type, allowing the fitted model to adjust for systematic differences across series, bake, flavor, appearance, and challenge/component categories. It also includes random intercepts for baker and episode.

Covariates and Predictors

HERE i USE tech scaled to $[-1, 1]$ within episodes to make components comparable!!!

The technical challenge requires special handling because it is originally recorded as a within-episode rank rather than an absolute judged component score. When technical performance is included, it enters through the scaled technical rank, which is directionally comparable with the judged component scores: higher values represent stronger performance and lower values represent weaker performance. Therefore, the baker-level random intercept is best interpreted as an adjusted overall performance measure across signature, showstopper, and scaled technical performance, rather than as a pure measure of judged component-score ability.

For baker i in series s and episode e, the score model can be written as

$$Y_{ise} = X_{ise}^{\top}\beta + \theta_i + v_e + \varepsilon_{ise},$$

where Y_{ise} is the scaled score outcome and X_{ise} contains the fixed effect indicators for series and component type.

- $(1 \mid id)$ gives each baker a random intercept. I interpret $\theta_i \sim N(0, \sigma_\theta^2)$ as a static proxy for the baker's adjusted performance level. Higher θ_i means baker i tends to receive higher scores, after adjusting for series, episode, and component type.
- Series is included as a fixed effect. This allows the model to adjust for systematic differences across seasons, such as contestant strength, scoring tendencies, judging standards, or challenge difficulty, without requiring estimation of a separate series-level variance component.
- $(1 \mid episode)$ gives each episode a random intercept. This captures episode-level residual scoring tendencies after adjustment for baker performance, series, component type, and other modeled structure. A negative episode effect means that bakers tended to receive lower scores than expected in that episode. This may reflect harder challenges, stricter judging, poor collective execution, or other episode-specific factors. Therefore, these effects should be interpreted as adjusted episode-level underperformance rather than direct measures of classically difficult challenges.
- Together, the random intercepts decompose the residual score tendency into baker-level and episode-level components. The total random-intercept contribution is

  $$\theta_i + v_e,$$

  where

  $$\theta_i \sim N(0, \sigma_\theta^2), \qquad v_e \sim N(0, \sigma_v^2), \qquad \varepsilon_{ise} \sim N(0, \sigma^2).$$

In matrix form, the fitted mixed model can be written as

$$Y = X\beta + Zb + \varepsilon,$$

where $X\beta$ is the fixed-effect part of the model, Zb is the random-effect part, and b contains the baker and episode random effects. The model assumes

$$b \sim N(0, G), \qquad \varepsilon \sim N(0, R),$$

so that the marginal covariance of the observed scores is

$$V = ZGZ^\top + R.$$

The fixed effects and variance components are estimated first, typically by restricted maximum likelihood. Once the covariance structure has been estimated, the fitted random effects are predicted from the fixed-effect residuals using

$$\hat{b} = \hat{G} Z^\top \hat{V}^{-1} (Y - X\hat{\beta}).$$

These fitted random effects are empirical-Bayes predictions, also called BLUP-style estimates in the linear mixed-model setting. They are not raw baker averages or unrestricted baker fixed effects. They are shrunken estimates: a baker's fitted random effect is pulled toward zero unless their observed residual performance is consistently above or below average. Bakers with fewer observations or noisier records are shrunk more strongly toward the population mean, while bakers with more consistent evidence are allowed to deviate more.

Equivalently, the fitted random effects can be understood as the solution to a penalized least-squares problem: the model tries to explain residual scoring patterns using baker and episode deviations, while penalizing deviations that are too large relative to the estimated between-baker and between-episode variance. This is the partial-pooling mechanism.

For prediction, the interpretation depends on whether the baker or episode was observed in the fitted data. For an existing baker, predictions can include that baker's fitted random effect, giving a subject-specific prediction:

$$\widehat{Y}_{ise} = X_{ise}^\top \hat{\beta} + \hat{\theta}_i + \hat{v}_e.$$

For a brand-new baker with no observed scores, there is no fitted baker-specific random effect yet. In that case, the natural population-level prediction sets the new baker effect to zero:

$$\hat{\theta}_{\text{new}} = 0.$$

If the new baker later accumulates observed scores, their random effect could be predicted using the same empirical-Bayes/BLUP logic: compare their observed scores with the fixed-effect prediction and shrink that residual information toward the population average according to the estimated covariance structure.

Thus, the fitted baker and episode random effects should be understood as shrunken model-based estimates of relative adjusted performance or episode-level scoring tendency, not as directly observed performance summaries. The resulting baker rankings are useful descriptive summaries of latent performance under this model, but they are not definitive measures of true intrinsic baking ability.

Baker Effects

baker_id	theta_hat	contestant	series	series_winner	eliminated_round
13.Jasmine	0.366	Jasmine	13	1	10
5.Sophie	0.333	Sophie	5	1	10
9.Giuseppe	0.322	Giuseppe	9	1	10
10.Syabira	0.303	Syabira	10	1	10
9.Crystelle	0.291	Crystelle	9	0	10
7.Steph	0.287	Steph	7	0	10
9.Jurgen	0.248	Jurgen	9	0	9
8.Peter	0.245	Peter	8	1	10
9.Chigs	0.245	Chigs	9	0	10
11.Josh	0.240	Josh	11	0	10
6.Rahul	0.231	Rahul	6	1	10
6.Kim-Joy	0.220	Kim-Joy	6	0	10
11.Tasha	0.210	Tasha	11	0	9
13.Tom	0.193	Tom	13	0	10
12.Gill	0.190	Gill	12	0	9
7.David	0.174	David	7	1	10
8.Dave	0.166	Dave	8	0	10
5.Steven	0.164	Steven	5	0	10
7.Alice	0.161	Alice	7	0	10
10.Maxy	0.155	Maxy	10	0	8
10.Sandro	0.151	Sandro	10	0	10
6.Dan	0.149	Dan	6	0	6
8.Lottie	0.137	Lottie	8	0	7
10.Abdul	0.114	Abdul	10	0	10
10.Janusz	0.111	Janusz	10	0	9
10.James	0.096	James	10	0	4
12.John	0.092	John	12	0	3
11.Cristy	0.089	Cristy	11	0	8
8.Mark	0.087	Mark	8	0	6
7.Michael	0.084	Michael	7	0	7
8.Hermine	0.081	Hermine	8	0	9
11.Dan	0.080	Dan	11	0	10

baker_id	theta_hat	contestant	series	series_winner	eliminated_round
12.Nelly	0.077	Nelly	12	0	6
12.Dylan	0.074	Dylan	12	0	10
12.Georgie	0.069	Georgie	12	1	10
6.Ruby	0.065	Ruby	6	0	10
13.Aaron	0.065	Aaron	13	0	10
12.Christiaan	0.063	Christiaan	12	0	10
7.Rosie	0.062	Rosie	7	0	9
5.Stacey	0.056	Stacey	5	0	9
7.Henry	0.050	Henry	7	0	8
5.Liam	0.049	Liam	5	0	8
11.Abbi	0.042	Abbi	11	0	3
11.Keith	0.038	Keith	11	0	2
12.Sumayah	0.029	Sumayah	12	0	7
13.Lesley	0.028	Lesley	13	0	7
8.Sura	0.025	Sura	8	0	4
7.Michelle	0.011	Michelle	7	0	5
5.Kate	0.007	Kate	5	0	10
5.Yan	0.002	Yan	5	0	7
9.Lizzie	0.001	Lizzie	9	0	8
11.Matty	-0.003	Matty	11	1	10
9.Freya	-0.004	Freya	9	0	5
8.Marc	-0.005	Marc	8	0	8
6.Antony	-0.008	Antony	6	0	3
12.Illiyin	-0.010	Illiyin	12	0	8
13.Jessika	-0.011	Jessika	13	0	4
13.Iain	-0.018	Iain	13	0	8
6.Jon	-0.019	Jon	6	0	7
6.Manon	-0.020	Manon	6	0	8
5.Flo	-0.022	Flo	5	0	3
5.Tom	-0.022	Tom	5	0	4
11.Nicky	-0.042	Nicky	11	0	5
13.Leighton	-0.046	Leighton	13	0	2
13.Toby	-0.053	Toby	13	0	9
5.James	-0.060	James	5	0	5
11.Rowan	-0.061	Rowan	11	0	5
7.Helena	-0.062	Helena	7	0	5
8.Laura	-0.066	Laura	8	0	10
7.Phil	-0.075	Phil	7	0	4
6.Briony	-0.075	Briony	6	0	9
9.Rochica	-0.084	Rochica	9	0	3
12.Andy	-0.084	Andy	12	0	5

baker_id	theta_hat	contestant	series	series_winner	eliminated_round
10.Maisam	-0.091	Maisam	10	0	2
10.Dawn	-0.093	Dawn	10	0	6
10.Kevin	-0.102	Kevin	10	0	7
12.Mike	-0.102	Mike	12	0	4
8.Linda	-0.107	Linda	8	0	5
13.Hassan	-0.109	Hassan	13	0	1
6.Imelda	-0.111	Imelda	6	0	1
6.Terry	-0.112	Terry	6	0	5
13.Nadia	-0.117	Nadia	13	0	5
11.Saku	-0.120	Saku	11	0	7
8.Mak	-0.121	Mak	8	0	2
13.Nataliia	-0.124	Nataliia	13	0	6
7.Priya	-0.125	Priya	7	0	6
5.Julia	-0.128	Julia	5	0	6
9.Jairzeno	-0.133	Jairzeno	9	0	2
7.Dan	-0.135	Dan	7	0	1
6.Karen	-0.149	Karen	6	0	5
9.George	-0.160	George	9	0	7
10.Carole	-0.162	Carole	10	0	5
6.Luke	-0.169	Luke	6	0	2
13.Pui Man	-0.173	Pui Man	13	0	3
5.Chris	-0.178	Chris	5	0	2
9.Tom	-0.189	Tom	9	0	1
8.Rowan	-0.194	Rowan	8	0	3
5.Peter	-0.202	Peter	5	0	1
7.Jamie	-0.202	Jamie	7	0	2
10.Will	-0.223	Will	10	0	1
11.Amos	-0.224	Amos	11	0	1
7.Amelia	-0.231	Amelia	7	0	3
8.Loriea	-0.247	Loriea	8	0	1
11.Dana	-0.250	Dana	11	0	6
10.Rebs	-0.260	Rebs	10	0	4
9.Amanda	-0.266	Amanda	9	0	6
9.Maggie	-0.269	Maggie	9	0	4
12.Hazel	-0.397	Hazel	12	0	2

series	series_effect_hat
12	0.201
9	0.109

series	series_effect_hat
13	0.108
8	0.096
11	0.066
7	0.051
10	0.048
6	0.047
5	0.000

these are less reliably estimated because there are relatively more of them compared to the fixed smaple size,

episode	episode_effect_hat	series	round
8.4	-0.107	8	4
7.8	-0.096	7	8
10.8	-0.086	10	8
12.5	-0.068	12	5
9.7	-0.067	9	7
8.8	-0.056	8	8
12.10	-0.056	12	10
9.4	-0.051	9	4
11.3	-0.051	11	3
6.8	-0.051	6	8
10.6	-0.049	10	6
11.4	-0.044	11	4
6.9	-0.044	6	9
11.7	-0.040	11	7
7.6	-0.039	7	6
6.5	-0.035	6	5
10.10	-0.034	10	10
9.1	-0.033	9	1
5.9	-0.032	5	9
5.3	-0.032	5	3
13.8	-0.030	13	8
5.8	-0.027	5	8
5.6	-0.026	5	6
10.4	-0.026	10	4
11.5	-0.024	11	5
13.4	-0.024	13	4
13.10	-0.023	13	10
11.10	-0.023	11	10

episode	episode_effect_hat	series	round
7.5	-0.021	7	5
9.10	-0.021	9	10
7.9	-0.020	7	9
6.6	-0.018	6	6
13.9	-0.018	13	9
8.7	-0.017	8	7
6.7	-0.016	6	7
11.8	-0.015	11	8
6.10	-0.015	6	10
7.3	-0.013	7	3
9.6	-0.011	9	6
5.1	-0.011	5	1
12.6	-0.006	12	6
13.2	-0.006	13	2
12.9	-0.006	12	9
7.10	-0.006	7	10
10.3	-0.004	10	3
12.7	0.001	12	7
7.2	0.004	7	2
13.1	0.004	13	1
13.6	0.006	13	6
12.8	0.006	12	8
13.3	0.008	13	3
11.9	0.008	11	9
5.10	0.010	5	10
8.10	0.010	8	10
10.5	0.014	10	5
5.4	0.014	5	4
11.6	0.016	11	6
8.3	0.016	8	3
6.3	0.016	6	3
10.9	0.019	10	9
12.1	0.019	12	1
9.8	0.021	9	8
8.2	0.022	8	2
12.4	0.026	12	4
8.6	0.027	8	6
9.5	0.028	9	5
10.7	0.030	10	7
9.3	0.030	9	3
8.9	0.030	8	9

episode	episode_effect_hat	series	round
12.2	0.030	12	2
7.7	0.031	7	7
8.5	0.032	8	5
5.5	0.033	5	5
13.7	0.033	13	7
5.7	0.035	5	7
5.2	0.036	5	2
9.9	0.037	9	9
10.1	0.037	10	1
8.1	0.042	8	1
7.1	0.047	7	1
6.1	0.048	6	1
13.5	0.049	13	5
6.4	0.049	6	4
12.3	0.053	12	3
11.1	0.060	11	1
6.2	0.064	6	2
9.2	0.068	9	2
10.2	0.098	10	2
11.2	0.112	11	2
7.4	0.112	7	4

Season Balance

The baker random effects can also be used to summarize how balanced each series was. Here, "balance" means how close the strongest adjusted bakers in a season were to one another under the mixed model. This is not a causal or definitive measure of cast strength. It is a descriptive summary of whether one baker clearly separated from the field, or whether the top of the season was tightly clustered.

The table below reports, for each series:

- the model's top-ranked baker;
- the winner and the winner's within-season rank;
- the strongest non-winner;
- the gap between the winner and the strongest non-winner;
- the gap between the top-ranked and third-ranked baker.

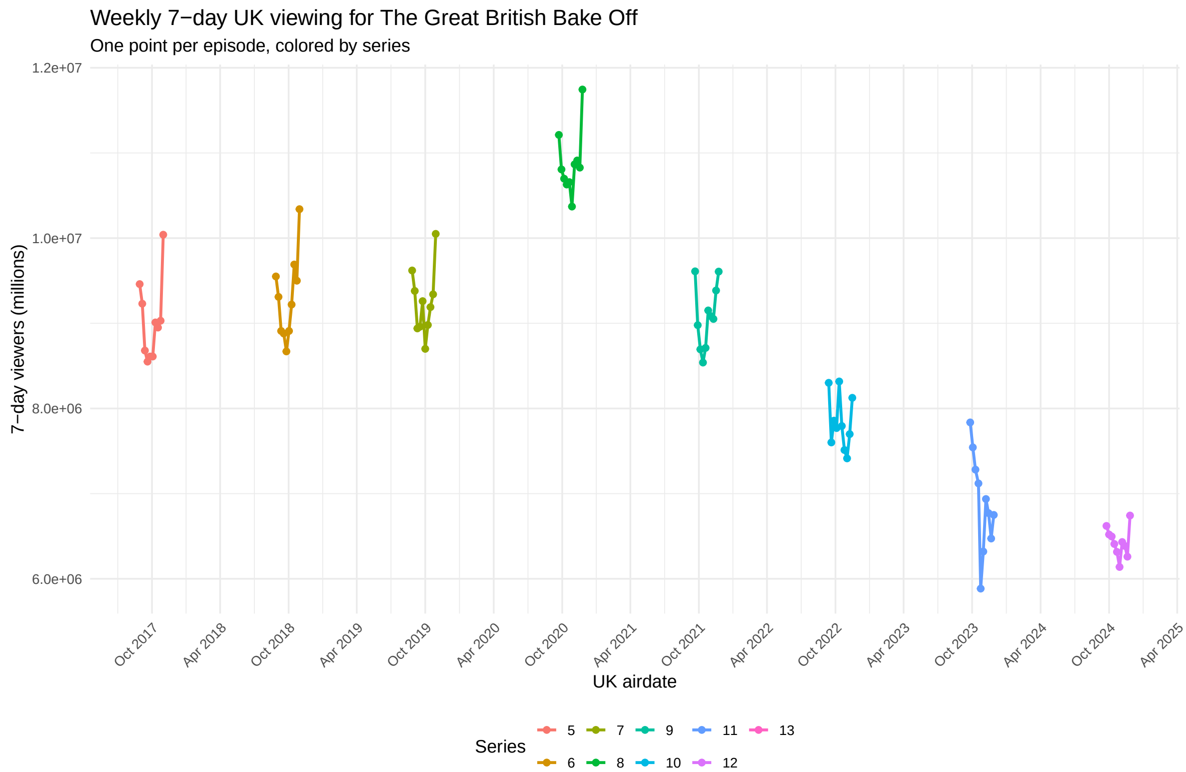
A large positive winner-versus-best-nonwinner gap suggests that the winner was clearly ahead of the rest of the field under this fitted model. A negative gap means the model ranks at least one non-winner above the winner. A small top-three spread suggests a more balanced season at the top.

Table 16: Season ba

Series	Winner	Winner Effect	Winner Rank	Best Non-Winner	Best Non-Winner Effect	Best Non-
9	Giuseppe	+0.322	1 of 12	Crystelle	+0.291	2
6	Rahul	+0.231	1 of 12	Kim-Joy	+0.220	2
8	Peter	+0.245	1 of 12	Dave	+0.166	2
12	Georgie	+0.069	5 of 11	Gill	+0.190	1
7	David	+0.174	2 of 13	Steph	+0.287	1
11	Matty	-0.003	7 of 12	Josh	+0.240	1
10	Syabira	+0.303	1 of 12	Maxy	+0.155	2
5	Sophie	+0.333	1 of 12	Steven	+0.164	2
13	Jasmine	+0.366	1 of 12	Tom	+0.193	2

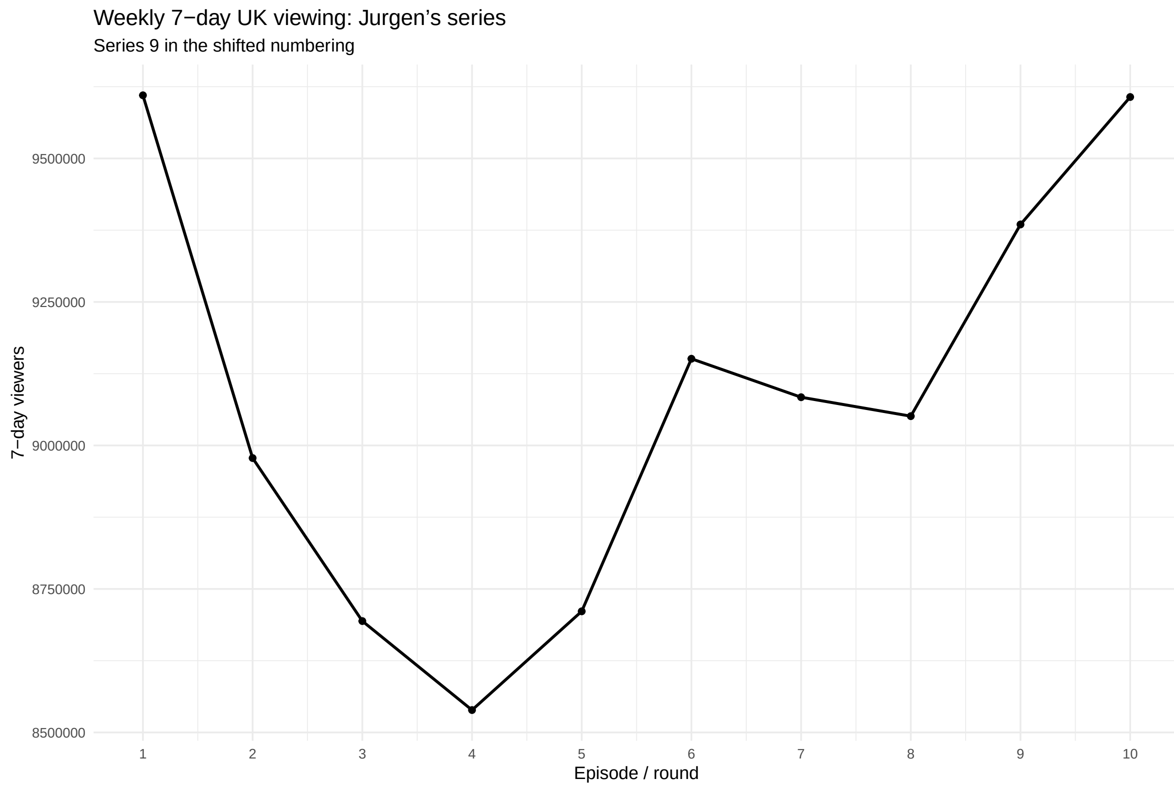
Viewers Analysis

Using the BARB 7 day viewing data

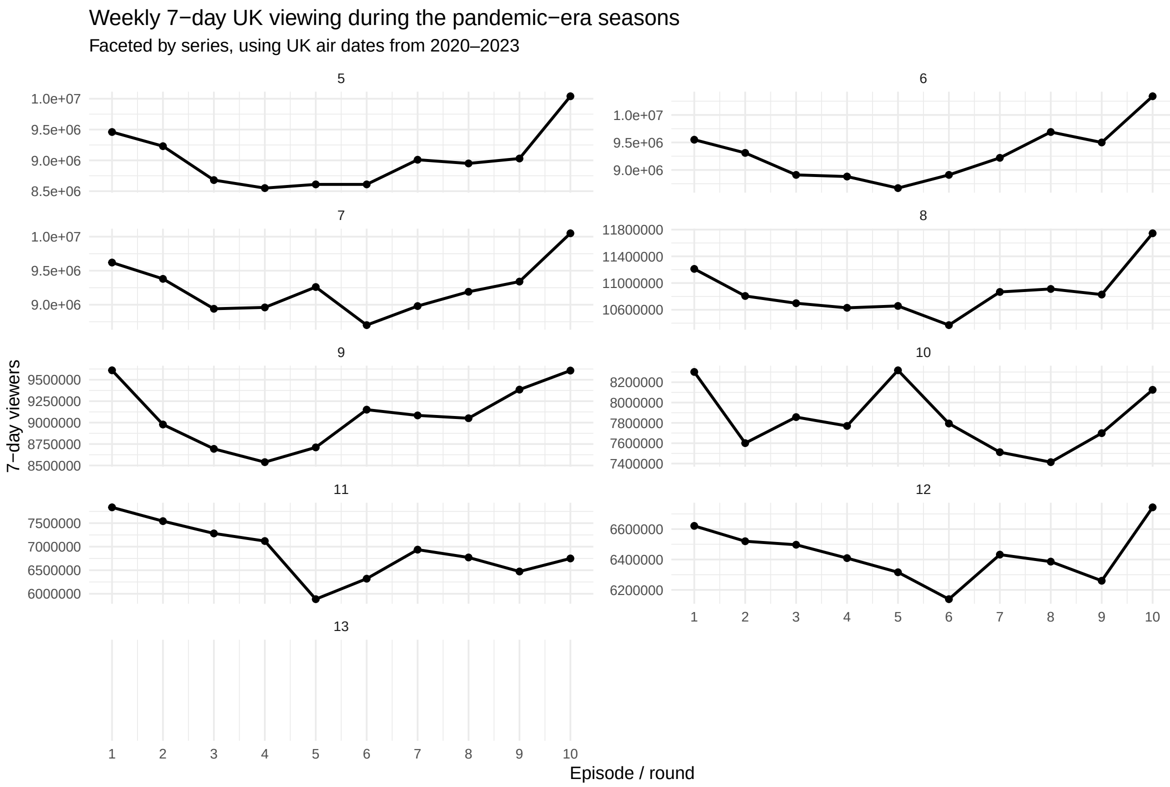


the pandemic jump is pretty clearly visaible here, letes look at season 9 and the pandemi seasons more clsoely

First series 9 for my beloved Jurgen,


Weekly 7−day UK viewing: Jurgen’s series
Series 9 in the shifted numbering
7−day viewers
9500000
9250000
9000000
8750000
8500000
1
2
3
4
5
6
7
8
9
10
Episode / round

facets,



Most seasons have a bit of a U shape - with more views in the begginng and end then the middle.

Overall Winning Probability Models

A natural way to track each baker's probability of winning is to use a sequential update. At the start of a series, before observing any performances, we have no baker-specific evidence and every contestant receives the same initial win probability:

$$P(W_i) = \frac{1}{n_c},$$

where W_i is the event that baker i wins the competition and n_c is the number of contestants in that series. In most series $n_c = 12$, although in the 2019 Netflix-numbered Series 7 there were 13 bakers.

After each round, we update these probabilities using the new round-level information. Let R_k denote the information observed in round k, such as signature performance, technical rank, showstopper performance, Star Baker status, elimination status, and the current risk set. In principle, a fully Bayesian update after round 1 would be

$$P(W_i \mid R_1) \propto P(R_1 \mid W_i)P(W_i),$$

and after rounds $1, \dots, k$,

$$P(W_i \mid R_1, \dots, R_k) \propto P(W_i) \prod_{r=1}^{k} P(R_r \mid R_1, \dots, R_{r-1}, W_i).$$

The hard part is specifying the likelihood contribution

$$P(R_k \mid R_1, \dots, R_{k-1}, W_i).$$

That would require a generative model for performances, judging decisions, elimination, and future survival. I do not try to build that full model here.

We could naively move forward and accumlate winning prob fro someone across the competition - however the final round is obviously weighted more than the other rounds in winning that is winning might be 60% performance in the actual final round and 40% past performance

Sequential Softmax Tracker

Instead, I use a simpler **sequential softmax tracker**. The idea is to summarize each baker's current evidence with a running strength signal, denoted s_{ik}. This signal may come from a sequentially fit mixed-effects performance model, a cumulative score summary, or another running measure of baker strength.

The update is

$$w_{ik} \propto w_{i,k-1} \exp(\lambda s_{ik}),$$

where $w_{i,k-1}$ is baker i's previous win-probability weight, s_{ik} is their current strength signal, and λ controls how strongly the current strength signal moves the probabilities.

After updating, the weights are normalized over the bakers still eligible to win:

$$P(W_i \mid R_1, \ldots, R_k) = \frac{w_{i,k-1} \exp(\lambda s_{ik})}{\sum_{j \in \mathcal{C}_k} w_{j,k-1} \exp(\lambda s_{jk})},$$

where $\mathcal{C}_k$ is the set of bakers still active after round k. Once a baker is eliminated, their probability of winning is set to zero.

Equivalently, the update is additive on the log-weight scale:

$$\log w_{ik} = \log w_{i,k-1} + \lambda s_{ik}.$$

So computationally this is a **sequential multiplicative-weights update followed by softmax normalization**. The previous win probabilities act like prior weights, and the current strength signal exponentially tilts those weights.

This is Bayesian in spirit: it starts from a prior, incorporates round-by-round evidence, and renormalizes to obtain updated win probabilities. But it is not a full Bayesian posterior unless $\exp(\lambda s_{ik})$ is explicitly defined as the likelihood contribution for round k. A more precise description is:

> a sequential Bayesian-style softmax tracker using a running mixed-model strength proxy as evidence.

The key constraint is that the strength signal must be estimated sequentially. We cannot use a mixed-effects model fit on the full data set, because that would leak future information into earlier rounds. For a genuine round-by-round tracker, the strength signal at round k must be estimated using only performances observed through round k, not later rounds or the final.

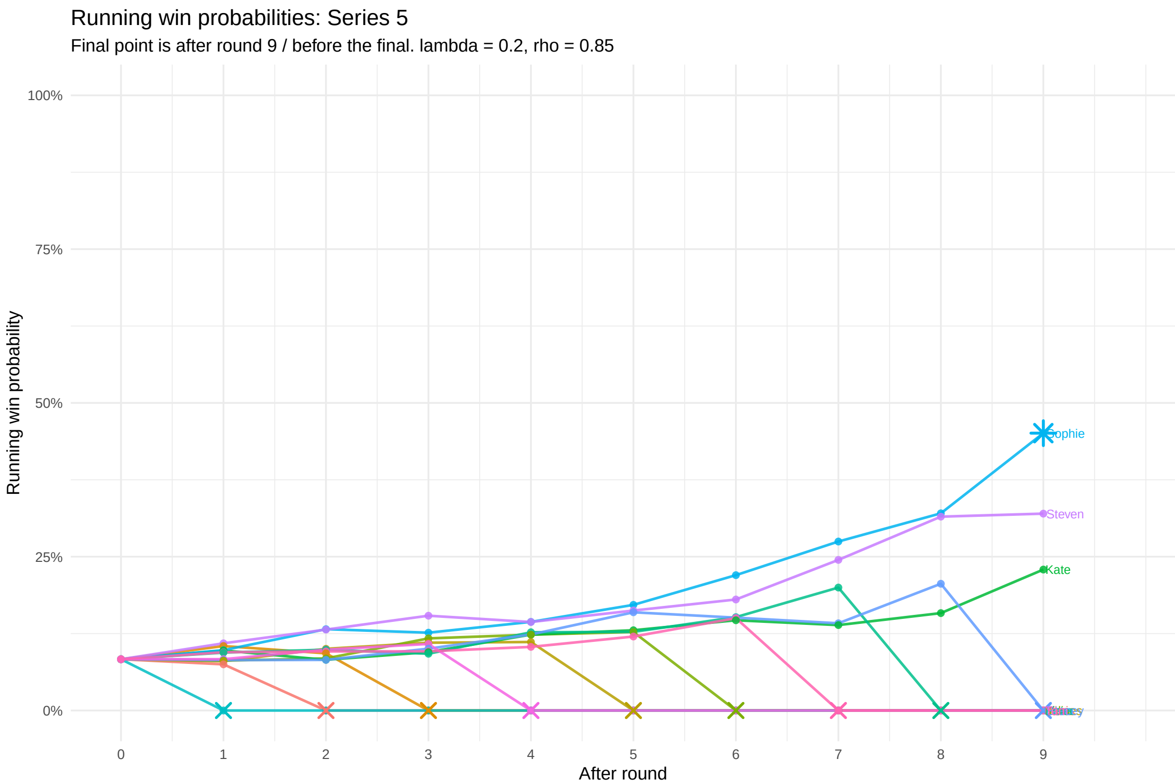
The tuning parameter λ is chosen by cross-validation using previous series data available up to that point. In other words, when predicting a given series, λ should be selected from earlier series rather than tuned using future information from the same series.

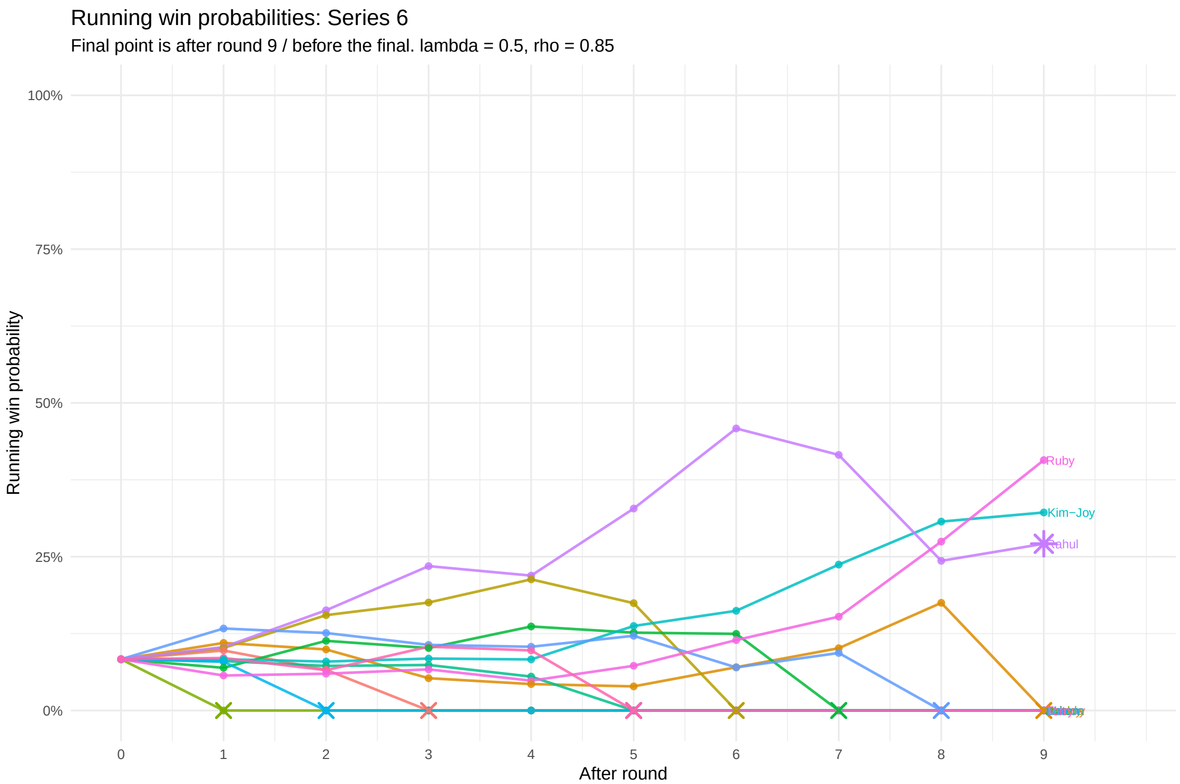
series	round	n_bakers	prob_sum	n_alive
5	0	12	1	12
5	1	12	1	11
5	2	12	1	10
5	3	12	1	9
5	4	12	1	8
5	5	12	1	7
5	6	12	1	6
5	7	12	1	5
5	8	12	1	4
5	9	12	1	3
5	10	12	1	1
6	0	12	1	12
6	1	12	1	11
6	2	12	1	10
6	3	12	1	9
6	4	12	1	9
6	5	12	1	7
6	6	12	1	6
6	7	12	1	5
6	8	12	1	4
6	9	12	1	3
6	10	12	1	1
7	0	13	1	13
7	1	13	1	12
7	2	13	1	11
7	3	13	1	10
7	4	13	1	9
7	5	13	1	7
7	6	13	1	6
7	7	13	1	5
7	8	13	1	4
7	9	13	1	3
7	10	13	1	1
8	0	12	1	12
8	1	12	1	11
8	2	12	1	10
8	3	12	1	9

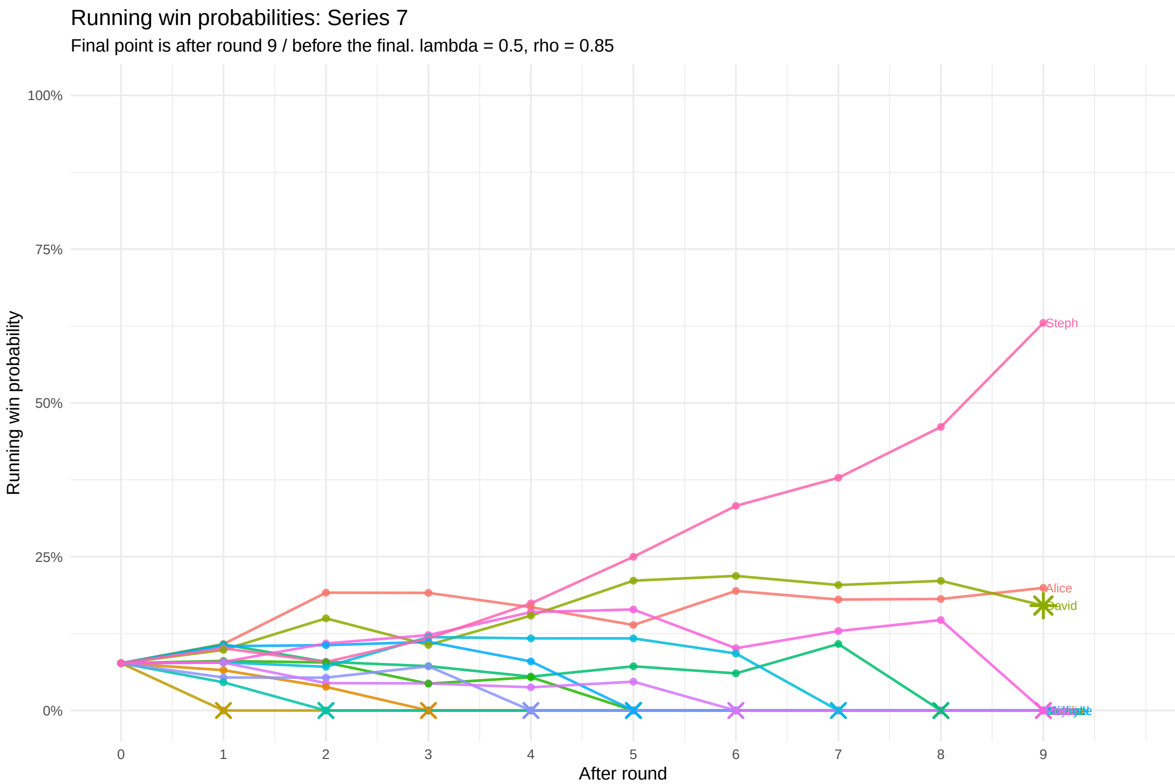
series	round	n__bakers	prob__sum	n__alive
8	4	12	1	8
8	5	12	1	7
8	6	12	1	6
8	7	12	1	5
8	8	12	1	4
8	9	12	1	3
8	10	12	1	1
9	0	12	1	12
9	1	12	1	11
9	2	12	1	10
9	3	12	1	9
9	4	12	1	8
9	5	12	1	7
9	6	12	1	6
9	7	12	1	5
9	8	12	1	4
9	9	12	1	3
9	10	12	1	1
10	0	12	1	12
10	1	12	1	11
10	2	12	1	10
10	3	12	1	10
10	4	12	1	8
10	5	12	1	7
10	6	12	1	6
10	7	12	1	5
10	8	12	1	4
10	9	12	1	3
10	10	12	1	1
11	0	12	1	12
11	1	12	1	11
11	2	12	1	10
11	3	12	1	9
11	4	12	1	9
11	5	12	1	7
11	6	12	1	6
11	7	12	1	5
11	8	12	1	4
11	9	12	1	3
11	10	12	1	1
12	0	11	1	11

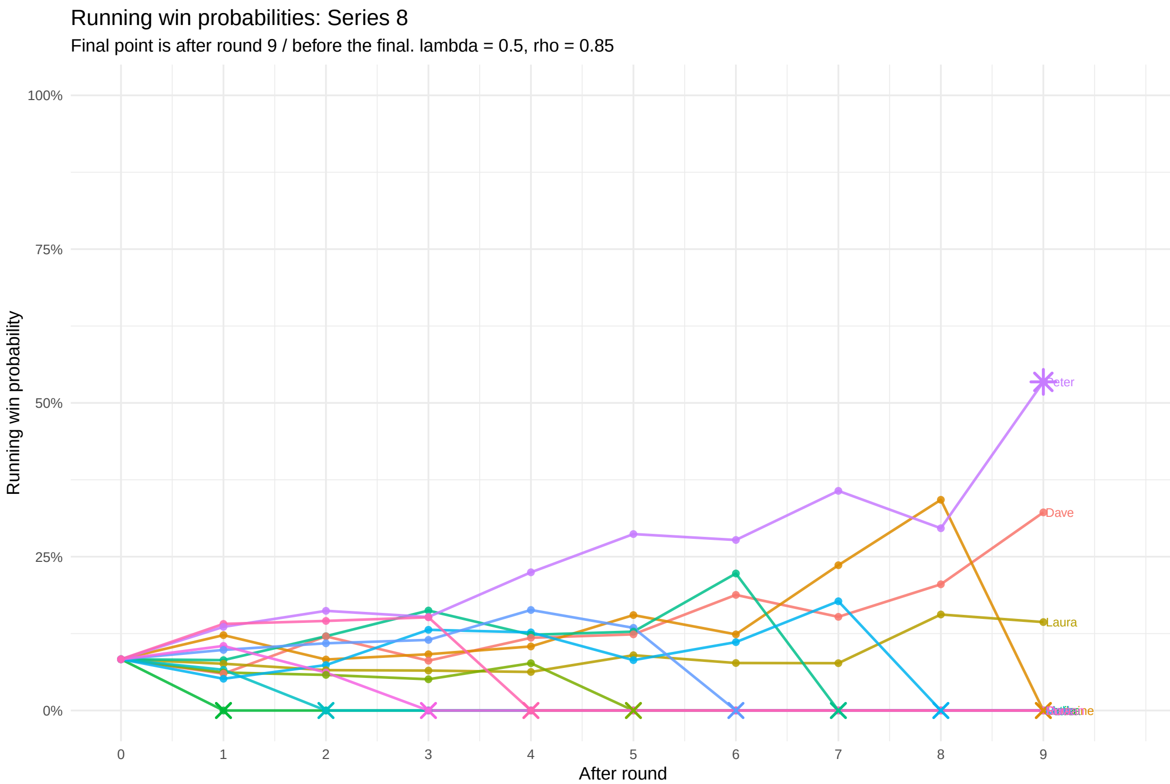
series	round	n_bakers	prob_sum	n_alive
12	1	11	1	11
12	2	11	1	10
12	3	11	1	9
12	4	11	1	8
12	5	11	1	7
12	6	11	1	6
12	7	11	1	5
12	8	11	1	4
12	9	11	1	3
12	10	11	1	1
13	0	12	1	12
13	1	12	1	11
13	2	12	1	10
13	3	12	1	9
13	4	12	1	8
13	5	12	1	7
13	6	12	1	6
13	7	12	1	5
13	8	12	1	4
13	9	12	1	3
13	10	12	1	1

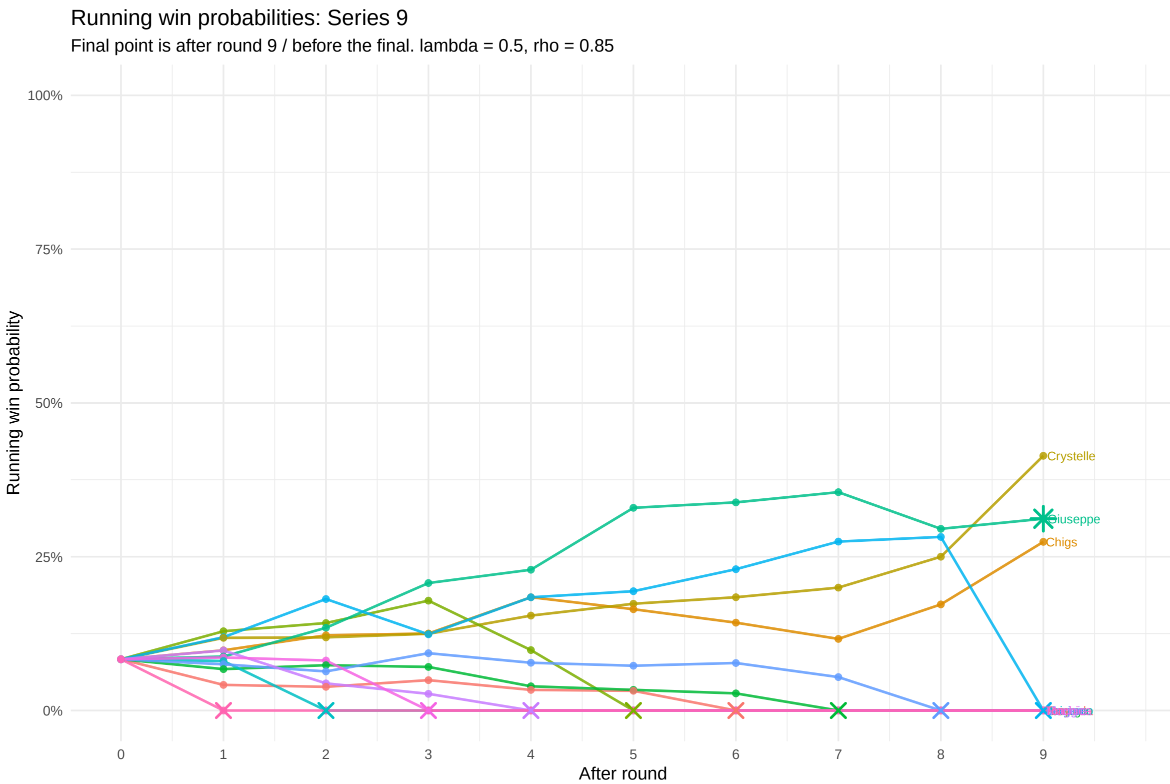
target_series	n_past_series	past_series	selected_lambda	selected_rho
5	0		0.2	0.85
6	1	5	0.5	0.85
7	2	5, 6	0.5	0.85
8	3	5, 6, 7	0.5	0.85
9	4	5, 6, 7, 8	0.5	0.85
10	5	5, 6, 7, 8, 9	0.5	0.85
11	6	5, 6, 7, 8, 9, 10	0.5	0.85
12	7	5, 6, 7, 8, 9, 10, 11	0.5	0.85
13	8	5, 6, 7, 8, 9, 10, 11, 12	0.5	0.85

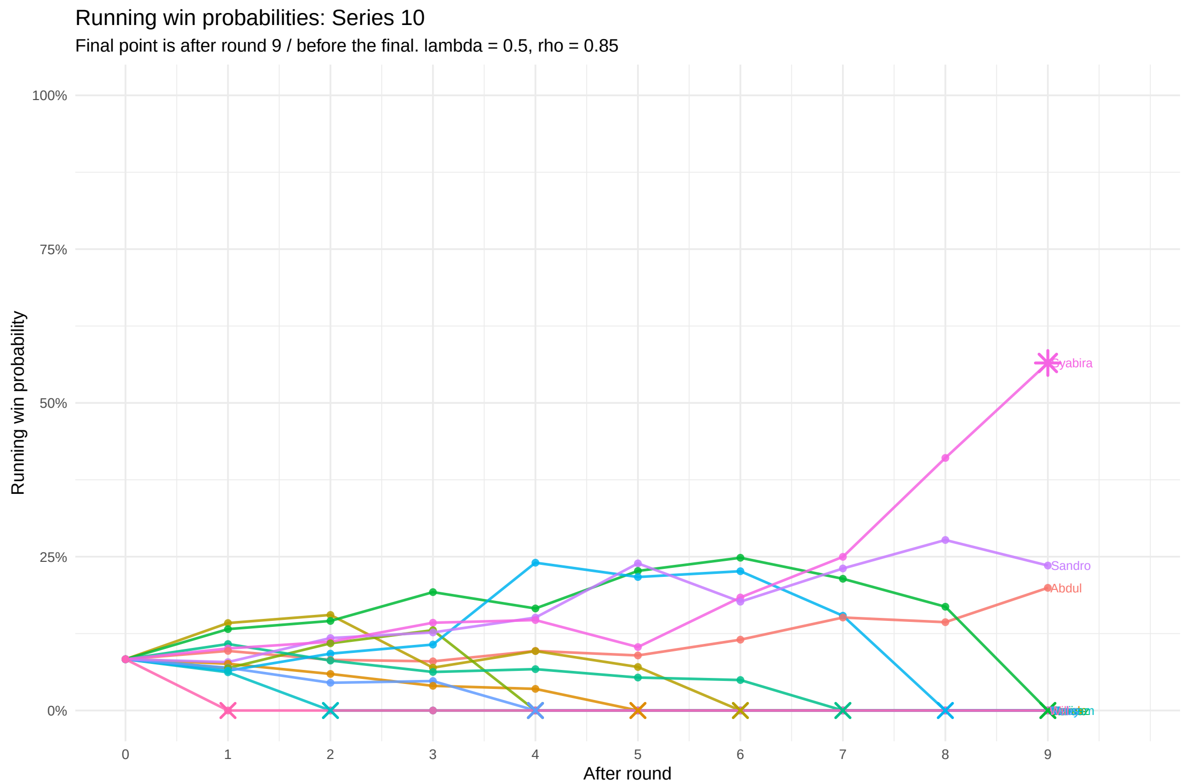

Running win probabilities: Series 5
Final point is after round 9 / before the final. lambda = 0.2, rho = 0.85
Running win probability
100%
75%
50%
25%
0%
Sophie
Steven
Kate
0
1
2
3
4
5
6
7
8
9
After round

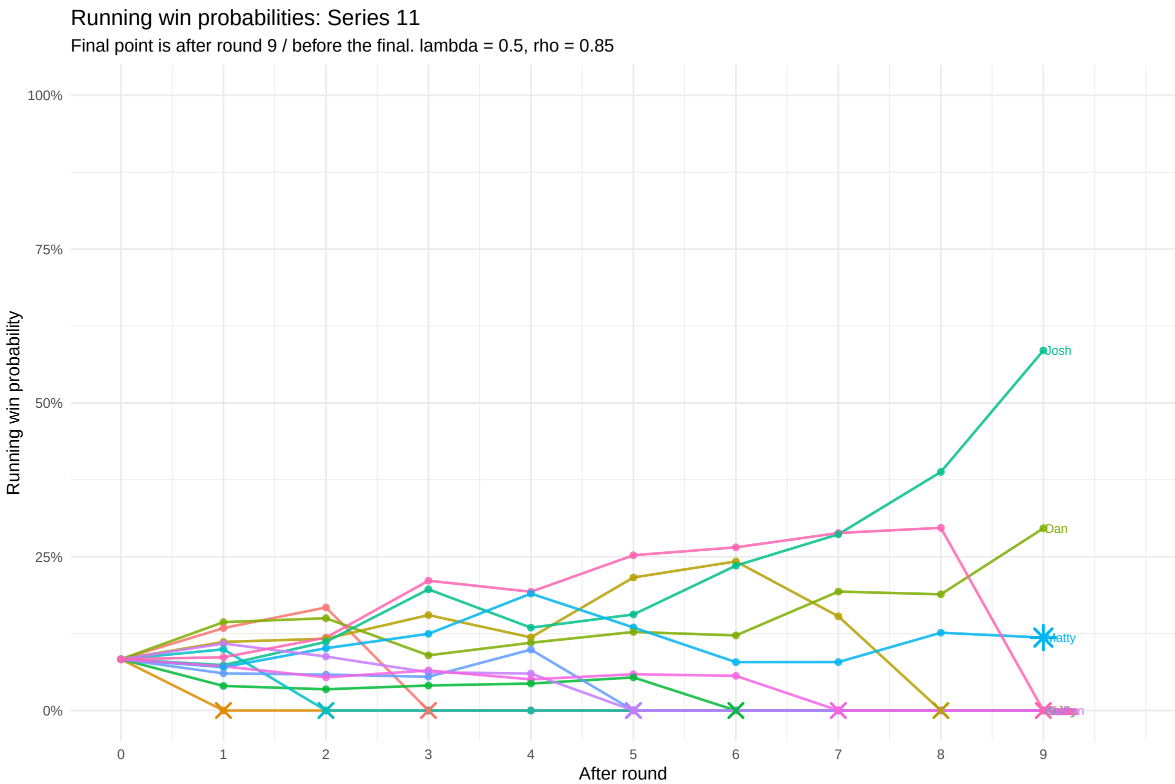

Running win probabilities: Series 6
Final point is after round 9 / before the final. lambda = 0.5, rho = 0.85
Running win probability
100%
75%
50%
25%
0%
Ruby
Kim−Joy
Rahul
0
1
2
3
4
5
6
7
8
9
After round

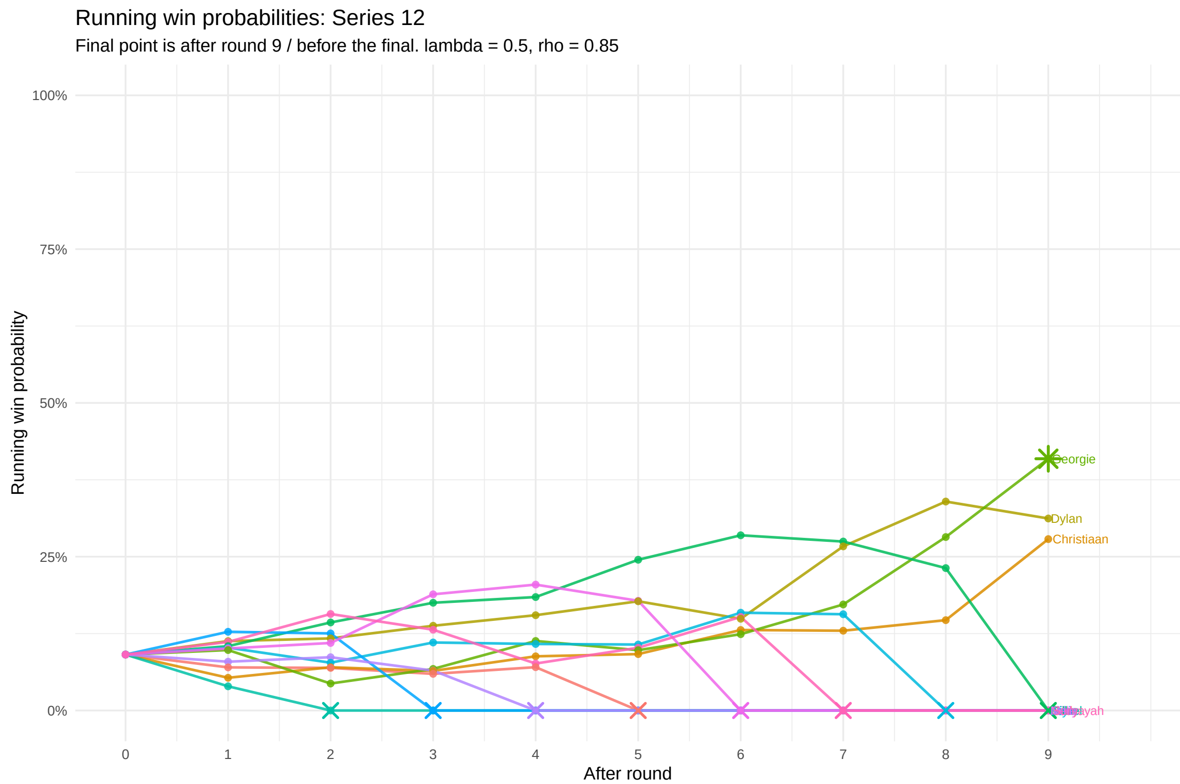

Running win probabilities: Series 7
Final point is after round 9 / before the final. lambda = 0.5, rho = 0.85
Running win probability
100%
75%
50%
25%
0%
0
1
2
3
4
5
6
7
8
9
After round
Steph
Alice
David

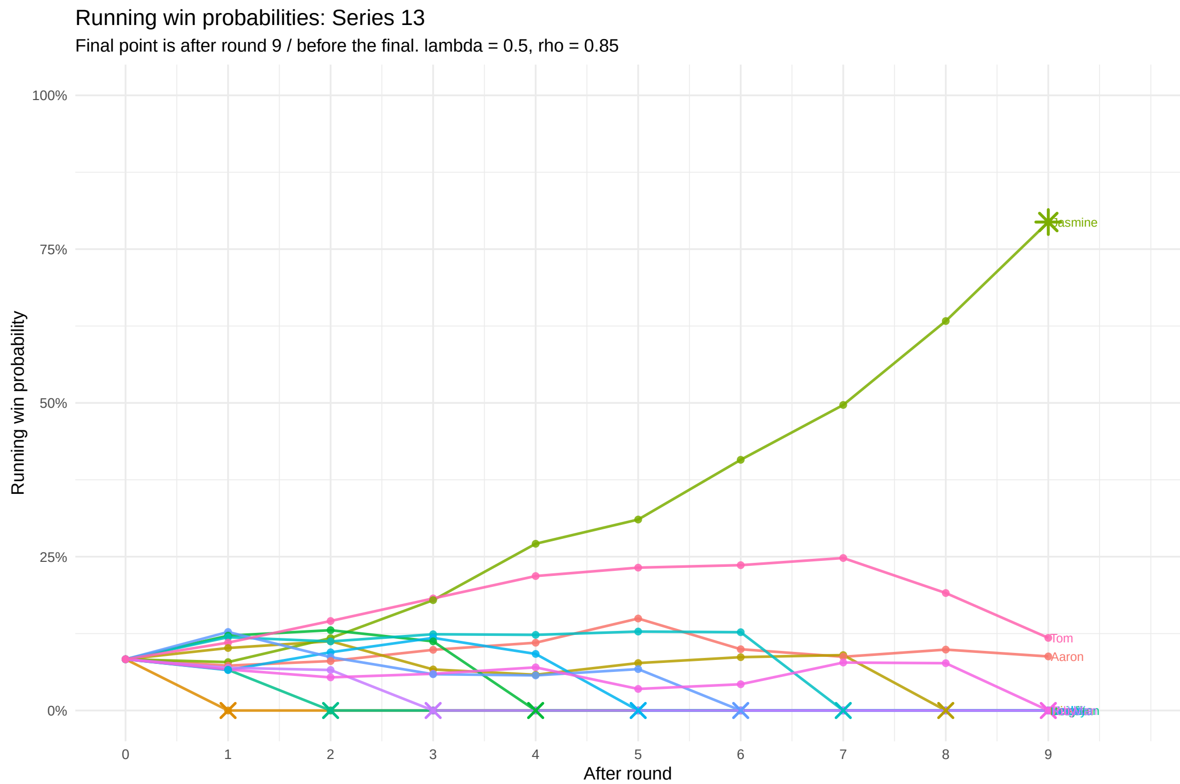

Running win probabilities: Series 8
Final point is after round 9 / before the final. lambda = 0.5, rho = 0.85
Running win probability
100%
75%
50%
25%
0%
After round
0
1
2
3
4
5
6
7
8
9
Peter
Dave
Laura


Running win probabilities: Series 9
Final point is after round 9 / before the final. lambda = 0.5, rho = 0.85
Running win probability
100%
75%
50%
25%
0%
0
1
2
3
4
5
6
7
8
9
After round
Crystelle
Giuseppe
Chigs


Running win probabilities: Series 10
Final point is after round 9 / before the final. lambda = 0.5, rho = 0.85
Running win probability
100%
75%
50%
25%
0%
Syabira
Sandro
Abdul
0
1
2
3
4
5
6
7
8
9
After round


Running win probabilities: Series 11
Final point is after round 9 / before the final. lambda = 0.5, rho = 0.85
Running win probability
100%
75%
50%
25%
0%
After round
0
1
2
3
4
5
6
7
8
9
Josh
Dan
Matty


Running win probabilities: Series 12
Final point is after round 9 / before the final. lambda = 0.5, rho = 0.85
Running win probability
100%
75%
50%
25%
0%
Georgie
Dylan
Christiaan
0
1
2
3
4
5
6
7
8
9
After round



These all show the overall winning probabilities as we move across rounds to round 9, just before the finale. The models failed to predict series 6, Rahul series 7 David and series 9 Giuseppes eventual win.

[fix to cv, it is naiver in sample now] Combining Star Baker Model and Overall Win Prob Model

We only built the sequential overall win probability model up to round 9 because the final episode that determines the series winner is determined by some weighting of performance up to the finale (from the sequential softmax model based on random effects) and performance in the actual finale. I want to estimate that weighting using a conditional logit model restricted to Round 10 finalists. This model is similar to the Star Baker model, but adds each baker's sequential win probability entering the finale as a covariate. It is fit using all the data from the finales first then we evaluate it using LOOCV (leave one finale out) and sequential CV of finales

series	episode	contestant

```
Call:
coxph(formula = Surv(rep(1, 27L), final_winner) ~ sig_sum_z +
    showstopper_sum_z + tech_rank_z + pre_final_logit_z + strata(episode),
    data = finale_model_data, method = "exact")

  n= 27, number of events= 9

                       coef exp(coef) se(coef)    z Pr(>|z|)
sig_sum_z         2.35e+01  1.56e+10 2.52e+03 0.01     0.99
showstopper_sum_z 5.15e+01  2.42e+22 3.06e+03 0.02     0.99
tech_rank_z       3.41e+00  3.03e+01 1.71e+03 0.00     1.00
pre_final_logit_z 2.24e+01  5.39e+09 2.04e+03 0.01     0.99

                  exp(coef) exp(-coef) lower .95 upper .95
sig_sum_z          1.56e+10   6.42e-11         0       Inf
showstopper_sum_z  2.42e+22   4.14e-23         0       Inf
tech_rank_z        3.03e+01   3.30e-02         0       Inf
pre_final_logit_z  5.39e+09   1.86e-10         0       Inf

Concordance= 1  (se = 0 )
Likelihood ratio test= 19.8  on 4 df,   p=6e-04
Wald test            = 0  on 4 df,   p=1
Score (logrank) test = 12.3  on 4 df,   p=0.02

        sig_sum_z showstopper_sum_z       tech_rank_z pre_final_logit_z
         1.56e+10          2.42e+22          3.03e+01          5.39e+09
```

Table 20: Approximate relative weighting in the finale model

component	coefficient	abs_coefficient	relative_weight
sig_sum_z	23.47	23.47	23.3%
showstopper_sum_z	51.54	51.54	51.1%
tech_rank_z	3.41	3.41	3.4%
pre_final_logit_z	22.41	22.41	22.2%

```
Call:
coxph(formula = Surv(rep(1, 27L), final_winner) ~ finale_lp_z +
    pre_final_logit_z + strata(episode), data = finale_model_data,
    method = "exact")

  n= 27, number of events= 9
```

```
                     coef exp(coef) se(coef)    z Pr(>|z|)
finale_lp_z          7.07   1172.76     6.54 1.08     0.28
pre_final_logit_z    1.25      3.48     1.08 1.16     0.25

                  exp(coef) exp(-coef) lower .95 upper .95
finale_lp_z         1172.76   0.000853   0.00315  4.37e+08
pre_final_logit_z      3.48   0.287361   0.42189  2.87e+01

Concordance= 0.889  (se = 0.101 )
Likelihood ratio test= 13.6  on 2 df,   p=0.001
Wald test            = 1.41  on 2 df,   p=0.5
Score (logrank) test = 8.82  on 2 df,   p=0.01

      finale_lp_z pre_final_logit_z
          1172.76              3.48
```

Table 21: Estimated relative weighting of pre-finale strength and finale performance

component	coefficient	abs_coefficient	relative_weight
Round 10 finale performance	7.07	7.07	85.0%
Pre-finale sequential win probability	1.25	1.25	15.0%

LOOCV

Table 22: Leave-one-finale-out predictions from the finale two-score model

series	episode	predicted_winner	winner	winner_rank	correct
5	5.10	Sophie	Sophie	1	TRUE
6	6.10	Rahul	Rahul	1	TRUE
7	7.10	David	David	1	TRUE
8	8.10	Peter	Peter	1	TRUE
9	9.10	Chigs	Giuseppe	2	FALSE
10	10.10	Syabira	Syabira	1	TRUE
11	11.10	Josh	Matty	2	FALSE
12	12.10	Georgie	Georgie	1	TRUE
13	13.10	Jasmine	Jasmine	1	TRUE

so with this combination of models an using LOOCV we predict the winner 7/9 times, improving just a slight bit over simply using the sequential model (6/9 final winners predicted) or using the star baker model in the finale (also getting 6/9 correct on both LOOCV and sequential CV)

Sequential CV

This does worse than the overall judging models because we have so little data on finales mainly

series	episode	contestant

Table 24: Fully prospective sequential finale CV predictions by finalist

series	contestant	sig_sum	showstopper	tech_rank	peer_sum	final_win_z	final_logit	final_pred_prob	pred_rank	winner_flag	n_train_finals	n_train_sb_episodes
8	Peter	3	3	2	53.4%	0.89	0.97	100.0%	1	Actual + predicted	3	36
8	Dave	2	1	1	32.2%	0.19	0.06	0.0%	2		3	36
8	Laura	-1	1	3	14.4%	-1.08	-1.03	0.0%	3		3	36
9	Chigs	1	3	2	27.4%	0.95	-0.84	100.0%	1	Predicted	4	45
9	Giuseppe	1	3	3	31.2%	0.10	-0.27	0.0%	2	Actual	4	45
9	Crystelle	0	1	1	41.4%	-1.05	1.11	0.0%	3		4	45
10	Syabira	3	2	2	56.5%	1.15	1.15	100.0%	1	Actual + predicted	5	54
10	Abdul	1	-2	1	19.9%	-0.59	-0.69	0.0%	2		5	54
10	Sandro	1	-1	3	23.6%	-0.56	-0.45	0.0%	2		5	54
11	Josh	3	0	1	58.5%	0.62	1.01	100.0%	1	Predicted	6	63
11	Matty	0	2	3	11.8%	0.53	-0.99	0.0%	2	Actual	6	63
11	Dan	-1	0	2	29.6%	-1.15	-0.02	0.0%	3		6	63
12	Georgie	0	3	2	40.9%	0.90	1.11	83.8%	1	Actual + predicted	7	72
12	Christiaan	2	0	1	27.9%	0.18	-0.82	15.0%	2		7	72
12	Dylan	-3	0	3	31.2%	-1.08	-0.29	1.2%	3		7	72

series	contestant	sig_sb	showstopper	tech_rank	semi_final_win_prob	final_tech_z	final_logit	final_prob	winner_rank	flag	train_n_finals	train_n_sb_episodes
13	Tom	1	3	1	11.8%	0.78	-0.50	63.2%	1	Predicted	8	81
13	Jasmine	1	3	3	79.4%	0.35	1.15	35.7%	2	Actual	8	81
13	Aaron	0	-1	2	8.8%	-1.13	-0.66	1.1%	3		8	81

Table 25: Fully prospective sequential finale CV: predicted versus actual winners

series	train_n_finals	train_n_sb_episodes	predicted_winner	predicted_prob	actual_winner	actual_prob	actual_rank	correct
8	3	36	Peter	NA	Peter	100.0%	1	Yes
9	4	45	Chigs	NA	Giuseppe	0.0%	2	No
10	5	54	Syabira	NA	Syabira	100.0%	1	Yes
11	6	63	Josh	NA	Matty	0.0%	2	No
12	7	72	Georgie	NA	Georgie	83.8%	1	Yes
13	8	81	Tom	NA	Jasmine	35.7%	2	No

Table 26: Fully prospective sequential finale CV summary

n_test_finals	n_correct	accuracy	median_actual_rank	mean_actual_prob
6	3	50.0%	1.5	0.533

Limitations

- The analysis is observational. The models estimate associations between performance and judging outcomes, not causal effects.
- The data depend on score-coding decisions. A different cleaned version of the data could lead to slightly different estimates. Thanks to Nathan Giusti (who in the linked performed their own analysis based on simple logistic regression - this ignores the round structure and risk set changing that the conditional logit handles) for preparing the underlying BakeOff data.
- Technical performance is originally a rank, not an absolute score. The scaled technical variable makes ranks more comparable across episodes, but it remains a transformed relative measure. In particular, it preserves within-episode ordering but does not turn technical rank into an absolute measure of technical quality.

- The conditional-logit Star Baker and elimination models correctly compare bakers within the episode-specific risk set. That is, they know who was actually competing each week. However, they do not explicitly model the fact that survival to later rounds is itself informative about latent ability. A baker appearing in round 8 is not exchangeable with a baker appearing only in round 1; survival is already a selection process.
- The prediction evaluation is cross-validated by episode, not by series. It tests whether the model generalizes to held-out episodes within the same data set, not whether it generalizes to a completely unseen GBBO season. If season-level effects are strong, such as different judges, different standards, or era-specific scoring patterns, the model may partially learn series-level structure that would not generalize cleanly to a new season. A stricter future evaluation would use leave-one-series-out cross-validation or genuinely held-out newer seasons.
- The latent baker-ability model treats each baker's random intercept as static. This summarizes each baker with a single average adjusted performance effect and ignores possible improvement, fatigue, adaptation, pressure effects, or round-specific trajectories over the course of the competition.
- A more flexible latent-performance model could include baker-specific round trends, such as random slopes for round by baker or baker-by-round interactions. This would allow ability to evolve over the season rather than being summarized by a single average performance effect. I did not use that model here because it would add complexity to an already limited data set.
- The baker random effects should be interpreted cautiously because later-round observations come from a selected sample of stronger bakers. Finalists are competing against other bakers who have already survived earlier rounds, so late-episode scores may compress estimated ability differences among strong bakers relative to bakers eliminated earlier. In other words, the model estimates adjusted scoring tendency from the observed competition path, not a fully selection-adjusted latent ability parameter.
- Episode effects are not direct challenge-difficulty scores. They may reflect difficulty, judging harshness, collective underperformance, unusual conditions, or other episode-specific factors. Negative episode effects are best interpreted as adjusted episode-level underperformance after accounting for baker, series, and component type, not as definitive evidence that a challenge was objectively harder.

References

- Nathan Giusti for preparing the underlying BakeOff data
- BARB for viewership data (I collected it via writing it down and trtanscribing)
- Agresti, A. Categorical Data Analysis, 3rd ed., Chapter 7, Section 7.3.